■ Eigensystem

Eigensystem[*m*] gives a list {*values*, *vectors*} of the eigenvalues and eigenvectors of the square matrix *m*.

Eigensystem finds numerical eigenvalues and eigenvectors if *m* contains approximate real numbers. ■ The elements of *m* can be complex. ■ All the eigenvectors given are independent. If the number of eigenvectors is equal to the number of eigenvalues, then corresponding eigenvalues and eigenvalues are given in corresponding positions in their respective lists. ■ The eigenvalues and eigenvectors satisfy the matrix equation m.Transpose[*vectors*] == Transpose[*vectors*].DiagonalMatrix[*values*]. ■ See page 452. ■ See also: NullSpace.

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